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PALAEOMAGNETIC AND POTASSIUM-ARGON DATING STUDIES OF THE TASMANIAN DOLERITES

By P. W. SCHMIDT & IAN McDOUGALL

(With 3 Tables and 2 Figures)

(MS received 2 June 1977)

ABSTRACT

The stable magnetizations of the Tasmanian Dolerites are shown to fall into two distinct groups depending upon their directions and the geographical region of the dolerites. It has been suggested that this could be a result of significant age differences between the dolerites of these groups. A series of K-Ar determinations indicates that there is no detectable systematic age differences and the average of the five bodies dated is 170.5 ± 8.0 m.y. (not significantly different from previous K-Ar dates from a single body). A re-appraisal of the palaeomagnetic data, in the light of the distinct groupings of the directions, yields two significantly different pole positions: Lat. 50.7°S, Long. 174.5°E (A_{95} = 5.2°) and Lat. 47.7°S, Long. 123.5° (A_{95} = 9.5°). The former of these is in excellent agreement with pole positions from other Lower to Middle Jurassic rocks of Australia but the significance of the latter remains obscure.

INTRODUCTION

The Tasmanian Dolerites occur as large intrusive sheets, commonly exceeding 300 m in thickness, and dyke-like bodies emplaced in nearly flat-lying Permian and Triassic sedimentary rocks; they crop out over an area of some 15 000 km² mainly in the eastern half of Tasmania (Fig. 1). Although stratigraphic control as to their age is poor the dolerites have traditionally been regarded as Jurassic, and this was confirmed by K-Ar dating of samples from the Red Hill Dyke in southern Tasmania (McDougall, 1961; 1962). The K-Ar ages on this intrusive body ranged from 140 to 170 m.y. based upon measurements on total-rock samples, plagioclase, anorthoclase, and clinopyroxene. McDougall (1961) suggested that the younger ages reflected loss of radiogenic argon, and he concluded that the best estimate of the age of emplacement was about 167 m.y., that is at about the Early-Middle Jurassic boundary.

Irving (1963) and Stott (1963) reported the results of palaeomagnetic measurements on the Tasmanian Dolerites; subsequently it was recognized that the palaeomagnetic pole position derived from these results was somewhat anomalous compared with Mesozoic pole positions from other continents when reassembled into Gondwanaland (Irving & *Robertson, 1969; Creer, Embleton & Valencio,* 1969; McElhinny & Embleton, 1974). This discrepancy is particularly striking when the results from the Tasmanian Dolerites are compared to those from the Ferrar Dolerites (Antarctica), which are considered to have been derived from a similar distinctive geochemical source (Compston *et al.*, 1968). Because of this Schmidt (1976) carried out a new palaeomagnetic study of the Tasmanian Dolerites,

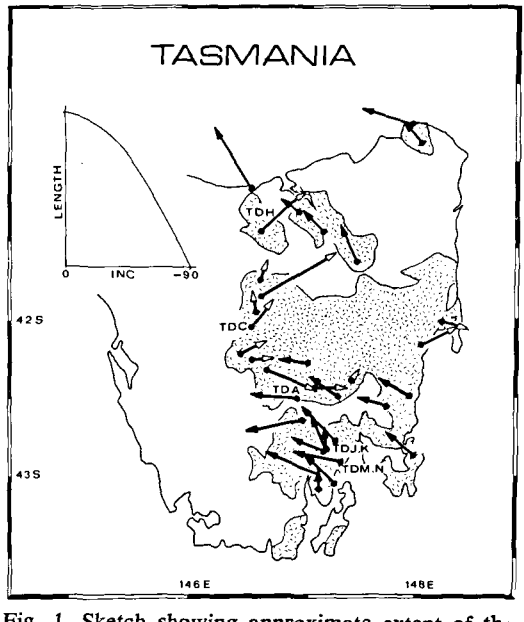


Fig. 1. Sketch showing approximate extent of the dolerites in relation to sampling localities and magnetizations. The pointers represent the magnetization at each sampling locality (see text). All directions are normal (*i.e.* negative inclinations) and the open and closed pointer heads have been used only to emphasize the relationship between the declination and geographical region. The sites from which samples have been dated are lettered (see Table 1).

and despite application of alternating field and thermal demagnetization procedures (to remove any secondary components of magnetization that may have been present) he found a palaeomagnetic pole (Lat. 52°S, Long. 156°E, A_{95} = 7°, N = 33) that is indistinguishable from that calculated from the data of Irving (1963) at Lat. 51°S, Long. 160°E (which was based only upon measurements of natural remanent magnetization). Schmidt (1976) noted, however, that the distribution of both the site mean-directions of magnetization and their virtual geomagnetic poles (VGP) are non-Fisherian, showing a spread in a latitudinal direction, He suggested that there may be two main groups of data which are derived from distinct geographic regions in Tasmania. Schmidt (1976) found it difficult to explain the distribution of virtual geomagnetic poles in terms of secular variation or tectonic movement subsequent to emplacement of the bodies, and suggested that the cause may be related to a significant spread in the age of emplacement and cooling of the dolerite bodies. The earlier age measurements were made on only one body (the Red Hill Dyke) whose VGP position, after reconstruction of Gondwanaland, conforms to Mesozoic pole positions from elsewhere, for example, Antarctica. To test the suggestion that the Tasmanian Dolerites were emplaced over a significant interval of time, a series of K-Ar age determinations on samples from several other intrusions distributed throughout Tasmania was undertaken.

METHODS

A number of samples were collected from dolerite bodies that represented both groups of palaeomagnetic directions. Additional oriented samples were collected from all sites (that were sampled for dating) for further palaeomagnetic studies. This was considered necessary to eliminate all doubts as to which data set a particular geochronological site represented. Samples for dating were collected also from the Red Hill Dyke in order to be able to directly relate the new results with the earlier K-Ar age measurements made by McDougall (1961).

All the samples were examined in thin section to determine their suitability for K-Ar age measurement. A number of samples were rejected because of extensive deuteric alteration, but several total-rock samples were considered sufficiently well crystallized to be used directly for dating. All samples are composed of plagioclase, clinopyroxene, iron oxide, and variable amounts of intersertal mesostasis made up of alkali feldspar and quartz. Several of the samples were considered to be too altered for measurement using a total-rock sample, but contained plagioclase that was free of alteration and of large enough grainsize to separate. In addition anorthoclase was separated from a granophyre from the Red Hill Dyke; this anorthoclase showed incipient alteration but was thought to be sufficiently fresh to make age measurements on it. In the earlier study McDougall (1961) used clinopyroxene from several samples for dating, but this approach was not followed in the present case because of doubts as to whether the measured potassium is actually contained within the pyroxene structure or is resident in small amounts of impurities in the separates. The separated minerals were better than 98 percent pure, and the total-rock samples were crushed to a fragment size of 1 to 0.5 mm.

The K-Ar age measurements were made by standard techniques (McDougall, 1966) using isotope dilution to measure argon and flame photometry to determine potassium. Measurements of the argon-isotope composition were made using a substantially modified AEI MS10 mass spectrometer with digital readout directly into a Hewlett-Packard 2116B computer, in which all data reduction was performed. The results of the age measurements are given in Table 1; the uncertainties in the reported ages are given at the level of one standard deviation derived from combining the errors from the measurement of isotope ratios, the tracer calibration, and the potassium measurement quadratically as described by McDougall *et al.* (1969). An average age is given for each sample, and the error assigned is either one standard deviation as calculated from the replicate ages or the standard deviation of the age that has the largest error, whichever is greater. This is done because in those cases where the replicates agree well, the calculated uncertainty for the mean age is unreasonably small when errors associated with potassium and tracer calibration measurements are taken into account.

Most palaeomagnetic samples were collected using a portable drill although a few block samples were taken which were later cored in the laboratory. Samples were oriented using both a magnetic compass and a sun compass (whenever possible). Three or four specimens were sliced from each core, the approximate dimensions of which are 2.5 cm (dia.) x 2.2 cm (height). All measurements and computing of directions were performed using a Digico spinner magnetometer (Molyneux, 1971) which has been improved by the addition of

Potassium-argon ages on samples from the Tasmanian Dolerites

Lab. No.	Field No.	Material	K (wt. %)	Rad. ^{40}Ar (10^{-10} mol/g)	100 Rad. ^{40}Ar / Total ^{40}Ar	Calculated Age (m.y.) ± 1 s.d.	Average Age (m.y.) ± 1 s.d.	Locality
76-176	TDN1	Total-rock	0.7992,0.7943	2.605 2.575	94.2 91.9	175.2 ± 1.5 173.3 ± 2.2	174.2 ± 2.2	Chilled dolerite, Snug River, Red Hill Dyke, Lat. 43.2°S, Long. 147.3°E
76-177	TDM5	Plagioclase	0.3304,0.3325	1.022 1.020	86.4 54.2	165.7 ± 1.5 165.3 ± 1.8	165.5 ± 1.8	Granophyre, Poverty Gully, Red Hill Dyke Lat. 43.0°S, Long. 147.2°E.
76-177	TDM5	Anorthoclase	5.020,4.946	14.452 14.726 14.726	90.1 86.3 84.0	156.2 ± 2.0 159.1 ± 2.1 159.1 ± 2.1	158.1 ± 2.0	
76-227	TDK6	Total-rock	0.7160,0.7170	2.341 2.205 2.225	94.8 94.9 63.7	175.1 ± 1.6 165.4 ± 1.5 166.8 ± 1.8	169.1 ± 5.2	Dolerites, adjacent to upper contact of sheet, southern outskirts of Hobart on Southern Expressway Lat. 42.9°S, Long. 147.3°E
76-232	TDJ5	Total-rock	0.6760,0.6643	2.202 2.206	95.7 94.1	176.1 ± 2.6 176.4 ± 2.6	176.3 ± 2.6	
76-231	TDA1	Plagioclase	0.3437,0.3396	1.157 1.127	91.7 91.9	181.0 ± 2.1 176.8 ± 2.0	178.9 ± 3.0	Dolerite, 3 km north of Cretna Lat. 42.7°S, Long. 146.9°E
76-228	TDC1	Plagioclase	0.1543,0.1540	0.505 0.507	56.1 42.8	175.6 ± 1.9 176.1 ± 2.0	175.8 ± 1.9	Dolerite, Butlers Gorge Lat. 42.3°S, Long. 146.3°E.
76-229	TDH3	Plagioclase	0.3132,0.3175	0.954 0.956	91.3 86.2	163.1 ± 2.6 163.5 ± 2.7	163.3 ± 2.6	Dolerites, 3 km east of Sassafrass, 16 km southeast of Devonport Lat. 41.3°S, Long. 146.5°E.
76-230	TDH4	Total-rock	0.4326,0.4288	1.327 1.289 1.262	95.8 85.1 82.9	165.6 ± 1.6 161.0 ± 2.0 157.8 ± 1.6	161.5 ± 3.9	
76-230	TDH4	Plagioclase	0.2903,0.2902	0.817 0.826	57.6 60.1	151.8 ± 1.4 153.4 ± 1.6	152.6 ± 1.6	

$\lambda_e = 0.585 \times 10^{-10}\ y^{-1}$ $\lambda_\beta = 4.72 \times 10^{-10}\ y^{-1}$ $^{40}K/K = 1.19 \times 10^{-2}$ atom %

Helmholtz coils above the μ-metal shield on the spinner unit. This was necessary because during the measurement procedure a specimen is exposed to the Earth's field in such a way that the resultant VRM (viscous remanent magnetization) is non-zero and when the NRM (natural remanent magnetization) is demagnetized sufficiently this extraneous component becomes appreciable. In addition to the Earth's field, a large vertical magnetic field is produced as a result of the geometry of the shield in the region where the specimen is exposed. The coils rectify this problem. AF (alternating field) demagnetization using the apparatus described by McElhinny (1966) was carried out to ensure that soft viscous magnetic components were eliminated. Statistical tests developed by Watson & Irving (1957) have been used to show, beyond reasonable doubt, that the distribution of the cleaned magnetic directions from the Tasmanian Dolerites, departs from that expected from a Fisher (1953) model.

RESULTS

The K-Ar results listed in Table 1 are from five intrusive bodies from localities distributed throughout the outcrop of the Tasmanian Dolerites (Fig. 1). Each measurement was replicated, and these results indicate that the reproducibility in general is better than about two percent. The overall spread in the measured K-Ar ages is from 152.6 m.y. (76-230 plagioclase) to 178.9 m.y. (76-231 plagioclase), or about 16 percent. This large spread in apparent age clearly is well outside that expected from experimental error alone, and indicates that either there is a real spread in the age of emplacement of the dolerites or that the assumptions underlying the calculation of age are not fulfilled in all cases. Assumptions made in calculating a K-Ar age include closed-system behaviour and the absence of radiogenic argon in the rocks at the time of crystallization. These factors can be assessed by examination of data from the individual intrusions.

Two samples from the Red Hill Dyke were dated, a total-rock sample (76-176) from adjacent to the chilled margin which gives an age of 174.2 $\pm$ 1.5 m.y., and anorthoclase and plagioclase separates from a granophyre (76-177) which yields ages of 158.1 $\pm$ 1.7 m.y. and 165.5 $\pm$ 1.5 m.y. respectively. The mean age is 165.9 $\pm$ 8.1 m.y., but the spread is such that geological effects must be present. As there is no obvious correlation of low potassium content with high age for these samples we doubt that the explanation for the spread is related to problems of excess radiogenic argon present at the time of recrystallization. We suspect, but cannot prove, that the differences in age are the result of variable loss of radiogenic argon, most evident in the case of the slightly altered anorthoclase. As noted earlier, the previous measurements on the Red Hill Dyke yielded ages in the range 144 to 167 m.y. (McDougall, 1961); the greater age of 167 m.y. was preferred and the lower ages were explained as resulting from loss of radiogenic argon by diffusion. From the combined results we suggest that the best estimate of age of emplacement of the Red Hill Dyke is about 170 $\pm$ 5 m.y.

Two samples from near the upper margin of a dolerite sheet in a road cutting on the Southern Expressway, on the outskirts of Hobart, give virtually concordant whole rock ages of 169.1 $\pm$ 5.2 m.y. and 176.3 $\pm$ 2.6 m.y., averaging 172.7 $\pm$ 5.1 m.y. This locality is some 18 km northnortheast of the Red Hill Dyke, and is indistinguishable in apparent age.

A sample from near Gretna, some 46 km northwest of Hobart, yields an age measured on a plagioclase separate of 178.9 $\pm$ 3.0 m.y.

A similar age of 175.8 $\pm$ 1.9 m.y. is found on a plagioclase sample of dolerite collected from near Butlers Gorge, some 110 km northwest of Hobart.

Samples of dolerite collected from the same body about 16 km southeast of Devonport and approximately 190 km northnorthwest of Hobart yielded ages of 152.6 $\pm$ 1.6 m.y. on plagioclase (76-230) and 161.5 $\pm$ 3.9 m.y. on the total-rock (76-230), and 163.3 $\pm$ 2.6 m.y. on plagioclase (76-229). Possibly the plagioclase from 76-230 has lost some of its radiogenic argon by diffusion, and on this basis the average apparent age for the other two samples, which give concordant ages, at 162.4 m.y. might be taken as the best estimate of the age of emplacement of this body. Taken at face value this age would seem to be significantly younger than those found for the other bodies except possibly that determined for the Red Hill Dyke.

Thus except for the results from the dolerite body near Devonport, the preferred ages for each body form a reasonably concordant group. However, the spread in ages on samples from a single body, presumably as a result of geological effects such as variable leakage of radiogenic argon, makes it difficult to determine whether there is a real range in age of emplacement of the Tasmanian Dolerites. Assuming that the five bodies sampled are representative of the Tasmanian Dolerites as a whole we can state from these results that the emplacement occurred over an interval of time that is unlikely to exceed 20 m.y. We cannot

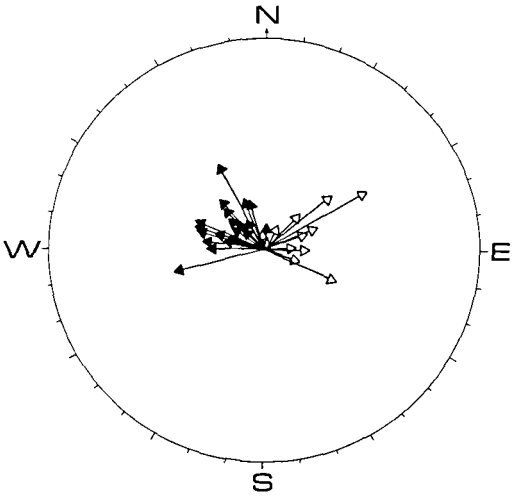


Fig. 2. The pointers represent the directions of magnetization from each site after transforming them to a common (mean) site to enable statistical tests (Table 2) to be applied. Note the distinct bimodal distribution.

exclude the possibility that emplacement occurred over a much shorter period of time.

Various estimates can be made as to the average age of the Tasmanian Dolerites. Giving each individual K-Ar age measurement listed in Table 1 unit weight a mean apparent age of 167.5 ± 8.8 m.y. is calculated. If the average age for each body is used, a mean of 170.5 ± 8.0 m.y. is obtained, and taking the oldest measured age from each of the five dolerite bodies (assuming that the lower ages reflect variable loss of radiogenic argon) a mean age of 173.7 ± 6.0 m.y. is found. All three averages are statistically indistinguishable from one another and indicate that emplacement occurred in the late Early Jurassic or Early-Middle Jurassic, as the boundary is estimated to have an age of about 172 m.y. (Harland *et al.*, 1964).

The palaeomagnetic results from the additional samples duplicate those reported by Schmidt (1976). The cleaned directions of magnetization measured here are indistinguishable from those found previously and, in general, differ by only a few degrees, which may be attributed to accumulated experimental errors. The site mean-directions are graphically depicted in Figure 1. The declination is the angle between the pointer and true north and the inclination can be found by comparing the length of the pointer to the corresponding inclination given by the inset in Figure 1. All magnetizations are directed upward (normal polarity) and the open and closed pointer heads have been used only to emphasize the correlation between the declination of magnetization and the site locality. Generally, the magnetizations in samples from the central highlands, and a small region on the east coast, are characterized by easterly declinations, whereas those from elsewhere are characterized by westerly declinations. It is quite evident from Figure 2 that the distribution is bimodal; the centres of the two populations being separated by about twenty degrees. The site mean directions have been transformed to their corresponding directions at a common (mean) site and tested statistically to determine quantitatively the degree to which the distribution departs from that expected of a circular distribution (Fisher, 1953). The results of two statistical tests are given in Table 2. First, these show that with 99% confidence the distribution is non-Fisherian and second, again at the 99% confidence level, that there are two distinct groups. It should be emphasized that the distribution is not split on an *ad hoc* basis, but rather on a geographical basis as mentioned above. It is ironical that the results here fail the statistical tests of Watson & Irving (1957), who used results from the Tasmanian Dolerites to illustrate their tests. This reflects the improvement in palaeomagnetic techniques that has been achieved since then.

From the new K-Ar results there does not seem to be any justification for correlating the different directions with different ages as previously suggested. There is no significant systematic age variation between the regions characterized by the different directions.

Other attempts to determine different characteristics of the two regions have failed. The appearance of the titanomagnetites in polished sections under reflected light show large variations in both regions, in grainsize and the extent of ilmenite exsolution. Other magnetic experiments, using saturation and back field techniques, reveal similar median coercivities of between 20 mT and 40 mT are present in specimens from both regions.

CONCLUSIONS

Perhaps the best overall estimate of age for the Tasmanian Dolerites is the mean of the average age for each of the five bodies dated, 170.5 ± 8.0 m.y. The internal scatter of results from individual bodies is such that we can only state that the Tasmanian Dolerites probably were emplaced over an interval of time of less than 20 m.y. The fact that the Tasmanian Dolerites all have normal magnetic polarity is of little assistance in limiting the time interval of emplacement because it

TABLE 2

Results of statistical tests

a. Chi-square Test

	Declination Ranges					
	1-60	61-120	121-180	181-240	241-300	301-360
Fo*	1	10	3	4	11	4
Fe*	5.5	5.5	5.5	5.5	5.5	5.5

*Fo - observed frequency, Fe - expected frequency, $\chi^2 = 14.82$; significance point with 99% confidence is 11.35

b. *F*-distribution Test (directions)

N_1	N_2	N	R_1	R_2	R	$F_{2,62}$	Significance*
21	12	33	20.834	11.833	32.334	62	Yes, at 99%

*Significance point with 99% confidence is 4.97

TABLE 3

Revised mean directions and pole positions for the Tasmanian Dolerites

Group	Direction					Pole		
	N	Dec(°)	Inc(°)	R	α_{95}	Lat(S)	Long(E)	A_{95}
East	12	62.8	-81.0	11.833	5.4	47.7	123.5	9.5
West	21	303.6	-79.2	20.834	2.9	50.7	174.5	5.2

appears that during the Early and Middle Jurassic the geomagnetic field was dominantly of normal polarity (McElhinny & Burek, 1971). The Tasmanian Dolerites are quartz normative tholeiitic basaltic rocks which show a most unusual trace element geochemistry in that K/Rb, Th/K, U/K, and initial $^{87}Sr/^{86}Sr$ ratios are more typical of rocks derived from a crustal environment than from the upper mantle (McDougall, 1962; Heier *et al.*, 1965; Compston *et al.*, 1968). The trace-element analyses were made on samples from the Red Hill Dyke and the Great Lake dolerite sheet, but there is no reason to doubt that this unusual geochemistry is characteristic of the province as a whole, particularly as the major-element chemistry and petrology of samples from many other bodies that have been examined are all very similar. These results argue strongly in favour of a single short-lived melting episode to produce these voluminous magmas, although it is difficult to make any firm estimate on the basis of such arguments as to the time involved in the production of the magmas and their emplacement in the upper crust.

The palaeomagnetic results may be accounted for in one of two ways. If it is assumed that the observations relate to original magnetic directions, then the dolerites have recorded an abnormally large geomagnetic field variation (*i.e.* an excursion). Also, although no reversed directions have been recorded, it is possible that the easterly directions are transitional from normal to reversed or *vice versa*. It is interesting to note that Creer *et al.* (1972) report a spread of directions from the South American Chon Aike lavas. Both normal and reversed directions were found. These lavas yield K-Ar ages (166 ± 5 m.y.) indistinguishable from those from the Tasmanian Dolerites which suggests a causative relationship for the spread of the directions. Alternatively, the directions may be composed of more than one component. Although a thermal overprint (perhaps related to the extrusion of Cainozoic basalts) was not revealed by thermal demagnetization, other components may still be present. For instance, it is conceivable that undetected low-temperature chemical alteration may have produced magnetic minerals whose blocking temperature and coercivity spectra overlap with those that carry the primary remanence. In this case thermal or AF demagnetization would not necessarily resolve the components.

Whatever the cause, the easterly directions are considered to be anomalous, in which case they should not be combined with the westerly group to give an overall mean direction for the Tasmanian Dolerites. It is clear that the most characteristic magnetic direction is the westerly direction which was found at 21 sites, as compared with 12 sites for the easterly directions. Results after separating these groups and recalculating the mean directions and mean pole positions are given in Table 3. The pole position from the westerly directions is in good agreement with poles from other Lower to Middle Jurassic rocks reported by Schmidt (1976). However, the origin and significance of the easterly directions remains, at present, inexplicable.

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